

FastAPI Practical – All Common Commands

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Objective

To install, create, run, test, and manage a FastAPI application using Python.

1. Check Python and pip

```
python --version
py --version
pip --version
```

2. Create a Virtual Environment

```
python -m venv venv
venv\Scripts\activate
```

3. Upgrade pip

```
python -m pip install --upgrade pip
```

4. Install FastAPI and Uvicorn

```
pip install fastapi uvicorn
```

5. Install Optional Useful Packages

```
pip install python-multipart
pip install pydantic
pip install httpx
pip install requests
```

6. Create the Application File

```
from fastapi import FastAPI
from pydantic import BaseModel

app = FastAPI(title='Usharani FastAPI Practical')

@app.get('/')
def home():
    return {'message': 'Hello, Usharani! FastAPI is running.'}

@app.get('/items/{item_id}')
def get_item(item_id: int):
    return {'item_id': item_id}

class Item(BaseModel):
    name: str
    price: float

@app.post('/items')
def create_item(item: Item):
    return {'item': item}
```

7. Run the FastAPI Application

```
uvicorn main:app --reload
uvicorn main:app --host 127.0.0.1 --port 8000
python -m uvicorn main:app --reload
```

8. Stop the Server

Press CTRL+C in the terminal.

9. Test in Browser

```
http://127.0.0.1:8000/
http://127.0.0.1:8000/docs
http://127.0.0.1:8000/redoc
```

10. API Testing with curl

```
curl http://127.0.0.1:8000/
curl http://127.0.0.1:8000/items/10
curl -X POST http://127.0.0.1:8000/items -H "Content-Type: application/json" -d
"{\"name\":\"Room\",\"price\":1500}"
```

11. Freeze and Install Dependencies

```
pip freeze
pip freeze > requirements.txt
pip install -r requirements.txt
```

12. Check Installed Packages

```
pip show fastapi
pip show uvicorn
pip list
```

13. Uninstall Packages

```
pip uninstall fastapi
pip uninstall uvicorn
```

14. Common Project Commands

```
mkdir fastapi_project
cd fastapi_project
code .
type nul > main.py
```

Result

The FastAPI application can be installed, created, executed, tested through the browser and automatic API documentation, and managed using common Python package commands.